STPM 2013 PHYSICS 960/3

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ *J s,* $c = 3.00 \times 10^{8}$ *m* s^{-1}, $e = 1.60 \times 10^{-19}$ *C,* $N_A = 6.02 \times 10^{23}$ mol^{-1}.

Section A [15 marks]

1. Which is **not** true of a simple harmonic motion?

A It is a periodic motion.
B Its motion involves a restoring force.
C The potential energy is always the same with its kinetic energy.
D The total energy of the motion is conserved.

Answer: C

Potential and kinetic energy interchange continuously; they are equal only at $x = A/\sqrt{2}$. Statement C is not true.

Explanation: SHM is periodic, has a restoring force and conserves total energy.

2. A particle undergoes a simple harmonic motion between points P and Q with O as its equilibrium position. If X is the midpoint between O and Q, what is the time taken for the particle to move from Q to X in terms of the period T?

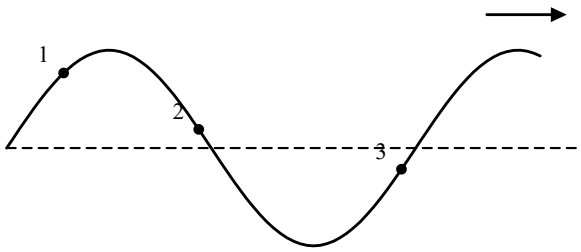
A $\frac{T}{12}$ B $\frac{T}{6}$ C $\frac{T}{4}$ D $\frac{T}{2}$

Answer: B

$$x = A\cos\omega t \text{ from } Q\text{: } \frac{A}{2} = A\cos\omega t \Longrightarrow \omega t = \frac{\pi}{3} \Longrightarrow t = \frac{T}{6}$$

Explanation: The particle moves slowly near the extreme, so covering the outer half takes longer than $T/8$.

3. A progressive wave in a stretched string which is travelling to the right is shown in the diagram below.



Which set of arrows shows the direction of motion of particles at points 1, 2 and 3 of the string?

	1	*2*	*3*
A	→	←	→
B	←	→	→
C	↓	↑	↓
D	↑	↓	↑

Answer: C

For a wave moving right, each point copies the motion of the point to its left: 1 (on the rising slope before the crest) moves down, 2 (on the falling slope) moves up, 3 (rising slope) moves down: ↓ ↑ ↓.

Explanation: Particles of a transverse wave move perpendicular to the direction of travel.

4. The superposition of two waves of equal amplitude and frequency but propagating in opposite directions at the same speed may result in

A a standing wave with the same frequency
B a standing wave with twice the frequency
C a progressive wave with the same frequency
D a progressive wave with twice the frequency

Answer: A

Two identical waves travelling in opposite directions superpose to give a standing wave of the same frequency.

Explanation: $y = 2A \cos kx \sin \omega t$ has the same ω.

5. What is the intensity level of the sound which is 3.0 m away from its point source of power 1.0×10^{-6} W?

[The threshold intensity is 1.0×10^{-12} W m^{-2}.]

A 39 dB B 45 dB C 50 dB D 55 dB

Answer: A

$$I = \frac{1.0 \times 10^{-6}}{4\pi(3.0)^2} = 8.8 \times 10^{-9} \text{ W m}^{-2}; \beta = 10 \log \frac{8.8 \times 10^{-9}}{10^{-12}} = 39 \text{ dB}$$

Explanation: Point source: the power spreads over $4\pi r^2$.

6. A car is travelling on a highway at a speed of 50 km h^{-1} while an ambulance with its siren on is approaching head-on. The actual frequency of the siren is 1.0 kHz. If the frequency heard in the car is 1.1 kHz, what is the speed of the ambulance?
[The speed of sound in the air is 330 m s^{-1}]

A 45 km h^{-1} B 55 km h^{-1} C 63 km h^{-1} D 150 km h^{-1}

Answer: C

$$v_o = 50 \text{ km h}^{-1} = 13.9 \text{ m s}^{-1}$$

$$1100 = 1000\frac{330 + 13.9}{330 - v_s} \Rightarrow v_s = 17.4 \text{ m s}^{-1} = 63 \text{ km h}^{-1}$$

Explanation: Both vehicles approach each other, so both effects raise the frequency.

7. When a baby is placed in front of a concave mirror with a radius of curvature R at a certain distance, an upright image which is twice the size of the baby is seen. What is the distance between the baby and the mirror?

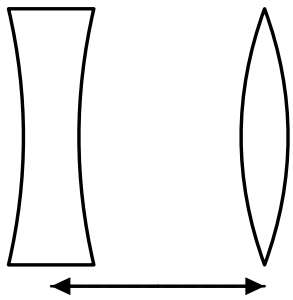
A $\frac{R}{4}$ B $\frac{R}{2}$ C $\frac{3R}{4}$ D R

Answer: A

$$f = \frac{R}{2}; \text{ upright, } m = 2: v = -2u \Rightarrow \frac{1}{u} - \frac{1}{2u} = \frac{1}{f} \Rightarrow u = \frac{f}{2} = \frac{R}{4}$$

Explanation: The baby must be inside the focal length.

8. The arrangement of a diverging lens and converging lens with a focal length of 10.0 cm each is shown in the diagram below.



The lenses arrangement is focused to a far object with the diverging lens in front. If the separation of the lenses is 10.0 cm, the image formed from the arrangement of the lenses is

A virtual image at 5.0 cm behind the converging lens
B virtual image at 10.0 cm behind the diverging lens
C real image at 10.0 cm behind the converging lens
D real image at 20.0 cm behind the converging lens

Answer: D

Diverging lens: a distant object gives a virtual image 10.0 cm in front of it.

This is 20.0 cm from the converging lens: $\frac{1}{v} = \frac{1}{10.0} - \frac{1}{20.0} \Longrightarrow v = 20.0$ cm (real, behind the converging lens)

Explanation: The image of the first lens is the object of the second.

9. The following statements are conditions for the superposition of two light waves to produce a sharp and distinct interference pattern **except**

A the waves have the same frequency
B the waves have constant phase difference
C the waves propagate in the same direction
D the waves have the same or almost the same amplitude

Answer: C

The waves do not need to propagate in the same direction to interfere; they must be coherent (same frequency, constant phase difference) and of similar amplitude. Option C is not a condition.

Explanation: Waves crossing at an angle still produce interference fringes.

10. In Young's two-slit experiment, the wavelength of the monochromatic light used is 6.0×10^{-7} m, and the separation of the slits is 0.40 mm. What is the angle made by the second-order maximum of the light?

A 3.8×10^{-3} rad B 3.0×10^{-3} rad C 1.5×10^{-3} rad D 7.5×10^{-4} rad

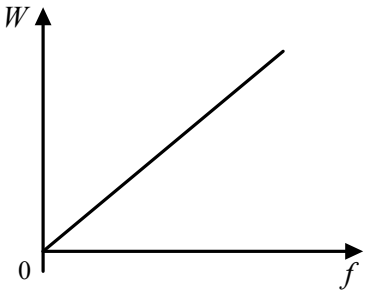
Answer: B

$$\sin\theta = \frac{2\lambda}{d} = \frac{2(6.0 \times 10^{-7})}{0.40 \times 10^{-3}} = 3.0 \times 10^{-3} \Rightarrow \theta = 3.0 \times 10^{-3} \text{ rad}$$

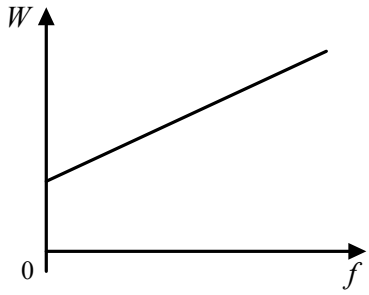
Explanation: For small angles, $\sin\theta \approx \theta$ in radians.

11. In a photoelectric effect experiment, the work function W of a metal is measured againts a frequency f of an incident light. Which graph shows the variation of the work function W with the frequency f?

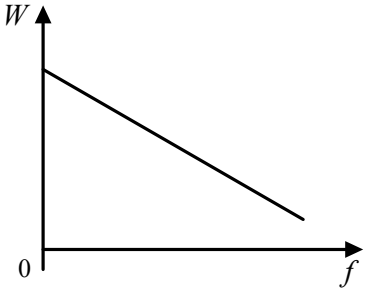
A



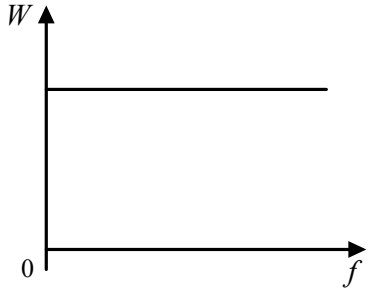
B



C



D



Answer: D

The work function is a property of the metal; it does not depend on the frequency of the light. A horizontal line (D).

Explanation: Frequency affects the kinetic energy of the photoelectrons, not the work function.

12. Which statement is true of the de Broglie wavelength?

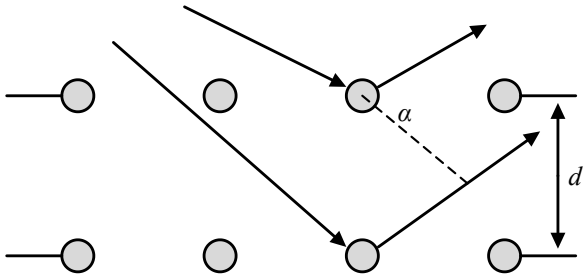
A A photon is associated with the wavelength.
B A particle with higher speed has a longer wavelength.
C A moving tennis ball has a wavelength in the range of a microwave.
D An electron can have a wavelength on the order of the size of an atom.

Answer: D

$\lambda = \frac{h}{mv}$: an electron (small mass) can have $\lambda \approx 10^{-10}$ m, the size of an atom

Explanation: Faster particles have shorter wavelengths; a tennis ball's wavelength is unimaginably small.

13. The diffraction of X-rays in a crystal is shown in the diagram below.



The maximum diffraction occurs only when the angle α between the incident ray and the reflected ray are 47.0° and 106.0° respectively. What are the possible values of the wavelength λ of the X-ray and the interatomic spacing d of the crystal?

	$\lambda/10^{-10}$ m	$d/10^{-10}$ m
A	1.00	1.25
B	1.25	1.00
C	1.00	2.17
D	2.17	1.00

Answer: A

The angle between the incident and reflected rays is 2θ: $\theta = 23.5°$ and 53.0°.

$\sin 53.0°/\sin 23.5° = 2.00$, so these are the 1st and 2nd orders: $2d \sin 23.5° = \lambda \Longrightarrow \lambda/d = 0.80$

Option A: $\lambda = 1.00 \times 10^{-10}$ m, $d = 1.25 \times 10^{-10}$ m.

Explanation: Bragg's law uses the glancing angle θ, half the deviation angle.

14. Which instrument can be used to measure the percentage abundance of the isotope of an element?

A Cloud chamber
B Mass spectrometer
C Electron microscope
D Geiger Muller counter

Answer: B

A mass spectrometer separates ions by mass, so it measures the relative abundance of isotopes.

Explanation: The other instruments detect radiation or image small objects.

15. The function of the moderator in a nuclear fission reactor is to

A absorb the excess neutrons in the reaction
B produce more neutrons in the chain reaction
C accelerate the neutrons produced in the reaction
D slow down the neutrons produced in the reaction

Answer: D

The moderator slows down the fast neutrons so that they are more likely to cause fission.

Explanation: Control rods absorb excess neutrons.

Section B [15 marks]

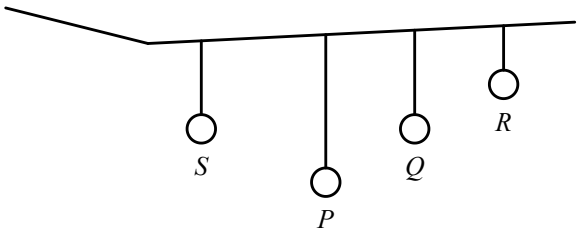
16. (a) State **two** differences between critical damping and over damping for an oscillatory system. [2]

Answer

Critical damping: the system returns to equilibrium in the shortest possible time; over damping: it returns much more slowly	1
Critical damping uses just enough damping to prevent oscillation; over damping has more damping than this	1
	2

Explanation: Neither oscillates.

(b) Three pendulums with identical light bobs *P*, *Q* and *R* which are of different lengths hang along a horizontally thin wire are shown in the diagram below.



Another pendulum with bob *S* which is heavy is hanged on the same wire and is set into an oscillation which triggers the other three pendulums to oscillate accordingly. Identify and give reasons which of the three bobs *P*, *Q* or *R* will have the highest amplitude. [3]

Answer

Q has the highest amplitude	1
Q has the same length as *S*, so its natural frequency equals the driving frequency of *S*	1
Resonance occurs: maximum energy transfer from *S* to *Q*	1
	3

Explanation: The natural frequency of a pendulum depends only on its length.

(c) A boy sitting on a swing is forced to oscillate by periodic pushes from his brother. Describe how resonance can be achieved. [2]

Answer

The pushes must have the same frequency as the natural frequency of the swing 1

and be applied in phase with the motion (e.g. at the same point in each swing) 1

2

Total for question **7**

Explanation: Each push then adds energy, so the amplitude builds up.

17. (a) Derive an expression for the number of radioactive nuclei N in a sample, at time t, in terms of the initial number of radioactive nuclei N_0 and a decay constant λ. [3]

Answer

$\frac{dN}{dt} = -\lambda N$ 1

$\int \frac{dN}{N} = -\lambda \int dt \Rightarrow \ln \frac{N}{N_0} = -\lambda t$ 1

$N = N_0 e^{-\lambda t}$ 1

3

Explanation: The rate of decay is proportional to the number of undecayed nuclei.

(b) Technetium-99 has a half-life of 6.00 hours. It is commonly used as a radioactive tracer in medicine. A fresh sample of 25.0 μg of technetium-99 is injected into a patient. Determine

(i) the decay constant of technetium-99, [2]

Answer

$\lambda = \frac{\ln 2}{T_{½}} = \frac{0.693}{6.00 \times 3600}$ 1

$\lambda = 3.21 \times 10^{-5}\ s^{-1}$ 1

2

Explanation: Convert hours to seconds.

(ii) the activity of the tracer after 30.0 minutes. [3]

Answer

$N_0 = \frac{25.0 \times 10^{-6}}{99} \times 6.02 \times 10^{23} = 1.52 \times 10^{17}$ 1

$N = N_0 e^{-\lambda(1800)} = 1.44 \times 10^{17}$ 1

$A = \lambda N = 4.6 \times 10^{12}$ Bq 1

3

Total for question **8**

Explanation: Molar mass of Tc-99 ≈ 99 g mol^{-1}.

Section C [30 marks]

18. (a)

(i) What is meant by *a transverse wave*? [1]

Answer

A wave in which the particles of the medium vibrate perpendicular to the direction of propagation 1

1

Explanation: Example: waves on a string, electromagnetic waves.

(ii) Distinguish between a transverse wave and a longitudinal wave. [1]

Answer

Transverse: vibrations perpendicular to the direction of travel; longitudinal: vibrations parallel to it (compressions and rarefactions) 1

1

Explanation: Sound in air is longitudinal.

(b) The periodic motion of a particle at the left end of a stretched string is defined by $y = 6.0 \sin \pi t$, where y is in centimetres and t in seconds. The wave disturbance travels along the string at a rate of 6.0 cm per second.

(i) Derive an expression for the displacement of a second particle on the string at point x to the right of the first particle at a later time t. [4]

Answer

The disturbance takes x/v to reach the particle at x	1
So its motion lags: $y = 6.0 \sin \pi(t - x/v)$	1
$v = 6.0$ cm s^{-1}	1
$y = 6.0 \sin \pi(t - \frac{x}{6.0})$ cm	1
	4

Explanation: This is a progressive wave travelling to the right.

(ii) Determine the displacement of the second particle if x is 80 cm and t is 1 minute. [1]

Answer

$y = 6.0 \sin \pi(60 - \frac{80}{6.0}) = 6.0 \sin (46.67\pi) = 5.2$ cm	1
	1

Explanation: 46.67π is equivalent to 0.67π (sin = 0.866).

(c) A transverse wave is given by $y = 6.0 \sin \pi(4.0t + 0.020x)$, where x and y are in centimetres and t in seconds.

(i) Determine the speed of the wave. [3]

Answer

$\omega = 4.0\pi$ rad s^{-1}, $k = 0.020\pi$ cm^{-1}	1
$v = \frac{\omega}{k} = \frac{4.0\pi}{0.020\pi}$	1
$v = 200$ cm s^{-1} (2.0 m s^{-1}) in the $-x$ direction	1
	3

Explanation: The plus sign between the terms means travel in the negative x direction.

(ii) Determine the maximum transverse speed of a particle of the wave. [3]

Answer

$v_{max} = A\omega$ 1

$= (6.0)(4.0\pi)$ 1

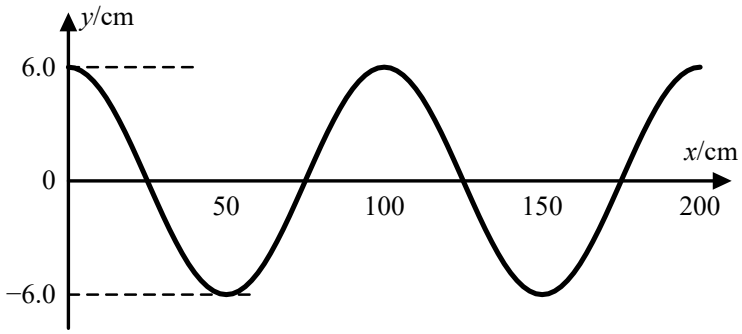
$= 75.4 \text{ cm s}^{-1}$ 1

3

Explanation: This is the particle speed, different from the wave speed.

(iii) Sketch the shape of the wave when t is 0.125 s. [2]

Answer



$y = 6.0 \sin (0.5\pi + 0.020\pi x) = 6.0 \cos (0.020\pi x)$: starts at a crest 1

Amplitude 6.0 cm and wavelength 100 cm shown 1

2

Total for question **15**

Explanation: $\lambda = 2\pi/k = 100$ cm.

19. (a)

(i) State Huygen's principle. [2]

Answer

Every point on a wavefront acts as a source of secondary wavelets 1

The new wavefront is the common tangent (envelope) to these wavelets 1

2

Explanation: The wavelets travel at the wave speed.

(ii) Draw a suitable diagram, and explain how a monochromatic light, after passing through Young's two-slit, produces two coherent light sources. [2]

Answer

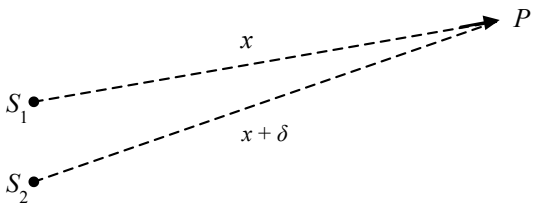
Light from a single source passes through a single slit; the same wavefront reaches both slits S_1 and S_2 — 1

By Huygens's principle each slit becomes a source of wavelets with the same frequency and a constant phase difference: coherent sources — 1

2

Explanation: Diagram: source → single slit → two slits equidistant from it → overlapping circular wavefronts.

(b) Waves from two coherent sources S_1 and S_2 meet at point P is shown in the diagram below.



The equations for the two waves are given by S_1: $E_1 = E_0 \sin(\omega t - kx)$ and S_2: $E_2 = E_0 \sin[\omega t - k(x + \delta)]$

(i) Derive the resultant wave equation E_R at point P. [2]

Answer

$E_R = E_1 + E_2 = E_0[\sin(\omega t - kx) + \sin(\omega t - kx - k\delta)]$ — 1

$E_R = 2E_0 \cos \frac{k\delta}{2} \sin(\omega t - kx - \frac{k\delta}{2})$ — 1

2

Explanation: Use $\sin A + \sin B = 2 \sin \frac{A+B}{2} \cos \frac{A-B}{2}$.

(ii) State the quantity in the wave equation that determines the intensity of the wave. [1]

Answer

The amplitude $2E_0 \cos(k\delta/2)$; intensity $\propto$ (amplitude)^2 — 1

1

Explanation: The amplitude depends on the path difference.

(iii) Deduce the values of the path difference δ in terms of the wavelength λ for the constructive interference and destructive interference. [4]

Answer

Constructive: $\lvert\cos(k\delta/2)\rvert = 1 \Longrightarrow k\delta/2 = n\pi$	1
$\delta = n\lambda$ $(n = 0, 1, 2, \ldots)$	1
Destructive: $\cos(k\delta/2) = 0 \Longrightarrow k\delta/2 = (n + ½)\pi$	1
$\delta = (n + ½)\lambda$	1
	4

Explanation: Use $k = 2\pi/\lambda$.

(c) A radio station operating at a frequency of 110 MHz has two identical antennae placed at the same level and 2.0 m apart. If a line passing through perpendicularly to the midpoint of the two antennae is in zero direction, at what angle can the first destructive interference be observed? [4]

Answer

$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8}{110 \times 10^6} = 2.73$ m	1
First destructive interference: path difference $d \sin\theta = \frac{\lambda}{2}$	1
$\sin\theta = \frac{2.73}{2(2.0)} = 0.682$	1
$\theta = 43.0°$	1
	4
Total for question	**15**

Explanation: The two antennae act as coherent sources in phase.

20. (a) Explain the terms *energy levels* in an atom, and state the reason for its negative signs. [2]

Answer

Energy levels are the discrete (quantised) values of energy that an electron in an atom can have	1
They are negative because zero is taken as the energy of a free electron at infinity; a bound electron needs energy to escape	1
	2

Explanation: The more negative the level, the more tightly bound the electron.

(b) Explain what do you understand by

(i) the ionisation energy of an atom, [2]

Answer

The minimum energy required to remove an electron completely from an atom in its ground state — 1

i.e. to move the electron from $n = 1$ to $n = \infty$ (13.6 eV for hydrogen) — 1

2

Explanation: It equals the magnitude of the ground-state energy.

(ii) the occurrence of a line spectrum. [2]

Answer

Electrons in excited atoms fall to lower energy levels, emitting photons of energy $hf = E_i - E_f$ — 1

Because the levels are discrete, only certain wavelengths are emitted, giving separate lines — 1

2

Explanation: Each line corresponds to one transition.

(c) A hydrogen atom in its second excited state undergoes a transition to the ground state by emitting photons. Determine the possible wavelengths of the emitted photons. [5]

Answer

Second excited state is $n = 3$; possible transitions: $3 \rightarrow 1$, $3 \rightarrow 2$, $2 \rightarrow 1$ — 1

$\lambda = \frac{hc}{\Delta E}$ — 1

$3 \rightarrow 1$: $\Delta E = 12.09$ eV, $\lambda = 103$ nm — 1

$3 \rightarrow 2$: $\Delta E = 1.89$ eV, $\lambda = 658$ nm (≈ 656 nm) — 1

$2 \rightarrow 1$: $\Delta E = 10.2$ eV, $\lambda = 122$ nm — 1

5

Explanation: An atom may return directly or in two steps, so three wavelengths are possible.

(d) A hydrogen atom is excited with its electron in the energy level $n = 4$. Determine the energy, in eV, that is required to ionise it? [2]

Answer

$E_4 = -\frac{13.6}{16} = -0.85$ eV — 1

Energy to ionise = 0.85 eV — 1

2

Explanation: Ionisation takes the electron from −0.85 eV to 0.

(e) An electron with energy 1.80×10^{-18} J collides with a hydrogen atom in its ground state and the electron is deflected. If the hydrogen atom is excited with its electron in the energy level $n = 2$, determine the energy of the deflected electron.

[The energy level for hydrogen atom is $\frac{-13.6}{n^2}$ eV, where n is an integer.] [2]

Answer

Incident energy = $\frac{1.80 \times 10^{-18}}{1.60 \times 10^{-19}}$ = 11.25 eV; excitation 1 → 2 needs 10.2 eV — 1

Energy of deflected electron = 11.25 − 10.2 = 1.05 eV (1.68×10^{-19} J) — 1

2

Total for question **15**

Explanation: An electron can transfer part of its energy in a collision, unlike a photon.